

Hanjin Seo

[Personal blog](#) | hseo@nd.edu | [Github](#)

RESEARCH INTERESTS

Research Topics	Microfluidics, protocells, giant unilamellar vesicles, complex coacervates, lipid nanoparticles, artificial cell membranes
Technical Disciplines	Microfluidic chip design and fabrication; droplet microfluidic-based giant unilamellar vesicle synthesis; enzyme reaction kinetic analysis within vesicles and coacervates; RNA-loaded lipid nanoparticle synthesis

EDUCATION

03/2019 – 02/2025	Pohang University of Science and Technology (POSTECH) Ph.D. in Chemical Engineering, advised by Prof. Hyomin Lee <i>Thesis: Spatiotemporal Control of Biochemical Reactions in Cell-Like Structures</i>	Pohang, Korea
03/2015 – 02/2019	Pohang University of Science and Technology (POSTECH) B.S. in Chemical Engineering, advised by Prof. Hyomin Lee	Pohang, Korea

TECHNICAL EXPERIENCES AND SKILL

Microscopy	Confocal fluorescence microscopy; optical diffraction tomography
EM	Cryo-electron microscopy (Cryo-EM); transmission electron microscopy (TEM); scanning electron microscopy (SEM)
Spectroscopy	UV-Vis spectroscopy; Raman spectroscopy; circular dichroism spectrometry (CD); Fourier transform infrared (FT-IR) spectroscopy
Particle Analysis	Dynamic light scattering (DLS); zeta potential analysis
Lithography	Photolithography
Programming	PyTorch, Matlab, Python, C

ACADEMIC EXPERIENCE

09/2025 – Present	University of Notre Dame Postdoctoral Researcher, advised by Prof. Matthew J. Webber • Lab automation and active machine learning-guided optimization of synthetic cell production	Indiana, USA
01/2025 – 08/2025	Pohang University of Science and Technology (POSTECH) Postdoctoral Research Associate, advised by Prof. Hyomin Lee	Pohang, Korea
09/2021 – 12/2021	Pohang University of Science and Technology (POSTECH) Lecturer, CHEB469H: ST-Career Exploration Seminar (undergraduate, 16 weeks)	Pohang, Korea
02/2019 – 01/2020	Pohang University of Science and Technology (POSTECH) Teaching Assistant, MECH250: Thermodynamics; CHEB417: Transport Phenomena I	Pohang, Korea

PUBLICATIONS

Lead author (7 papers)

- [1] L. Jeon, S. Kim, **H. Seo**, H. S. Lee, G.-S. Na, G.-R. Yi, and H. Lee, “3D-printed milli-fluidic mixing-extrusion platform for continuous, high-throughput LNP production,” *Small*, p. e74386, 2026. (Co-first authors: L. Jeon, S. Kim, and **H. Seo**.)
- [2] **H. Seo**, H. S. Lee, and H. Lee, “Salt bridge-mediated coacervate-to-vesicle transformation in arginine-rich coacervates,” *ACS Nano*, vol. 19, p. 16890, 2025.
- [3] **H. Seo** and H. Lee, “Programmable enzymatic reaction network in artificial cell-like polymersomes,” *Advanced Science*, vol. 11, p. 2305760, 2024.
- [4] **H. Seo**, L. Jeon, J. Kwon, and H. Lee, “High-precision synthesis of RNA-loaded lipid nanoparticles for biomedical applications,” *Advanced Healthcare Materials*, vol. 12, p. 2203033, 2023.

- [5] **H. Seo** and H. Lee, “Spatiotemporal control of signal-driven enzymatic reaction in artificial cell-like polymersomes,” *Nature Communications*, vol. 13, p. 4179, 2022.
- [6] **H. Seo** and H. Lee, “Recent developments in microfluidic synthesis of artificial cell-like polymersomes and liposomes for functional bioreactors,” *Biomicrofluidics*, vol. 15, p. 021301, 2021.
- [7] **H. Seo**, C. Nam, E. Kim, J. Son, and H. Lee, “Aqueous two-phase system (ATPS)-based polymersomes for particle isolation and separation,” *ACS Applied Materials & Interfaces*, vol. 12, p. 55467, 2020.

Co-author (1 paper)

- [8] L. Jeon, Y. Kim, J. Yoon, **H. Seo**, and H. Lee, “One-step synthesis of hydrogen-bonded microcapsules for pH-triggered protein release,” *Small Structures*, vol. 4, p. 2300200, 2023.

SELECTED PRESENTATIONS

- [1] **H. Seo** and H. Lee, “Microfluidic synthesis of polymersomes with semi-permeable membrane for spatiotemporal control of biochemical reactions,” oral presentation, Fall Meeting of the American Chemical Society (ACS), Chicago, USA, 2022.
- [2] **H. Seo**, C. Nam, and H. Lee, “Versatile vesicles-based artificial cells via droplet microfluidics,” poster presentation (Excellent Paper Award), Fall Meeting of the Polymer Society of Korea, Jeju, Korea, 2019.
- [3] C. Nam, **H. Seo**, E. Kim, and H. Lee, “Microfluidic generation of aqueous two-phase system (ATPS) droplets: Toward artificial cell and biocompatible encapsulation,” poster presentation (Excellent Paper Award), Fall Meeting of the Korean Society of Industrial and Engineering Chemistry, Jeju, Korea, 2018.

PATENTS

- [1] H. Lee, C. Nam, **H. Seo**, and E. Kim, “Aqueous two-phase system (ATPS)-based polymersomes for single-particle-level isolation and intensification,” 1024820100000, 2022.
- [2] J.-Y. Lim, G.-R. Yi, D.-S. Hwang, H. Lee, S. Park, **H. Seo**, L. Jeon, and S. Kim, “Chamber device for manufacturing lipid nanoparticles,” submitted.

HONORS AND AWARDS

2025	Graduate Student of the Year, Department of Chemical Engineering, POSTECH
2024	Soseon Scholarship - Science and Technology Field, Community Chest of Korea
2023	Soseon Scholarship - Science and Technology Field, Community Chest of Korea
2023	POSTECHIAN Fellowship, POSTECH
2022	POSTECHIAN Fellowship, POSTECH
2022	Donating POSTECHIANs Scholarship, POSTECHIAN Foundation
2022	Scholarship Recipient, Paan Scholarship and Culture Foundation
2019	Excellent Paper Award, The Polymer Society of Korea
2018	Excellent Paper Award, The Korean Society of Industrial and Engineering Chemistry